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Rasch Methods for Beginners



Why is knowledge of “Rasch methods” of interest to psychologists, policy makers, teachers, students and researches? Why might a basic overview of a psychometric method (Rasch) be of interest to such a diverse group of individuals? In a nutshell- if you are using tests or surveys in some manner (be it developing a test/survey, evaluating test/survey data, making decisions from test/survey results, or just interested in better interpreting the testing/surveying of respondents) Rasch techniques can help you improve the quality of your efforts. Generally, Rasch techniques are useful when evaluating data from multiple choice tests, partial credit tests, and rating scale Likert surveys- thus if that is the type of data you might collect or evaluate or make decisions based upon- Rasch can often be of an immediate help. Rasch techniques for tests and surveys have revolutionized how tests and surveys are developed and how data is evaluated and results communicated in many fields of research (e.g. Social Sciences, Medicine). It is the author’s view that regardless of one’s interests and goals, basic Rasch techniques are not difficult to master.

More specifically what can Rasch techniques allow you to do?

Rasch techniques can provide a guiding framework that will help you develop new tests and new surveys that address issues associated with reliability and validity.

Rasch techniques can furnish you with a roadmap of factors to consider if you are contemplating the use of existing tests/surveys for a project. Of course the roadmap can be used for tests and surveys you might wish to develop.

Rasch software can help you evaluate your data and compute so called “person measures” (e.g. a person’s overall test performance, a person’s overall attitude toward a set of survey items) and “item difficulties” (for example, which test items are most difficult, which survey items are easiest to agree with).

Rasch methods can be used to help anyone with an interest in using test and survey data to communicate findings to stakeholders in an understandable (nontechnical) manner.

What is provided to readers of this article? In this article the author provides a brief non-technical overview of some techniques which were pioneered by the Danish mathematician Georg Rasch (Rasch, 1960) and extended by the American Psychometrician Benjamin Wright (Wright and Stone, 1979, Wright and Masters, 1982).

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Some Problems With Commonly Utilized Techniques To Evaluate Test and Survey Data

In my years of work with Rasch techniques I have found the best way to approach explaining “Why Rasch” is accomplished by pointing out some of the deficits in how ratings scale data is evaluated, then to present a sample of Rasch techniques one can utilize. Below I describe two difficulties with commonly used techniques to evaluate survey data.

Rating scale data is ordinal: To best understand the shortfalls with commonly used techniques to evaluate rating scale data, let us consider what is typically done by someone evaluating a data set collected with 9 survey items and a 5-step rating scale. First off there would be some sort of coding used to code the respondent’s answers to each of the 9 survey items. One coding format might be to use a 1 for an answer of Strongly Disagree, a 2 for an answer of Disagree, a 3 for an answer of Neutral, a 4 for an answer of Agree and a 5 for an answer of Strongly Agree. Below a sample Excel sheet is provided with the answers and coding for 4 respondents who answered the nine-item survey. One can see that the first respondent answered a 3 (a Neu-

tral) for Q1 of the survey and one can see that this same respondent answered a 3 (a Neutral) to Q2 of the survey. One can also see that the second respondent in the data set answered a 5 (a Strongly Agree) for both Q8 and Q9. Column J presents the total raw score for each respondent (the nine answers added up for a respondent) and column K presents the average raw score for each respondent (total raw score/9).

What is the error in this spread sheet? What exactly is the problem with such calculations? First and foremost, rating scale data is ordinal data. That means that with rating scale data such as in this example all one knows is that Strongly Agree means more agreement than Agree, that Agree means more agreement than Neutral, that Neutral means more agreement than Disagree, and that Disagree means more agreement than Strongly Disagree. This means with the coding that we have used a $5 > 4 > 3 > 2 > 1$. This means we do not know the distance between Strongly Agree, Agree, Neutral, Disagree and Strongly Disagree. Figure 2 presents a schematic of different scenarios in which $5 \text{ (Strongly Agree)} > 4 \text{ (Agree)} > 3 \text{ (Neutral)} > 2 \text{ (Disagree)} > 1$

	A	B	C	D	E	F	G	H	I	J	K
1	Q1 (No 68)	Q2 (No 69)	Q3 (No 70)	Q4 (No 71)	Q5 (No 72)	Q6 (No 73)	Q8 (No 75)	Q9 (No 76)	Q10 (No 77)	Total Raw Score	Average Raw Score
2	3	3	3	3	4	4	3	3	3	29	3.222222222
3	3	5	4	4	4	5	5	5	5	40	4.444444444
4	2	4	4	3	3	3	1	1	2	23	2.555555556
5	4	4	3	3	5	5	4	5	5	38	4.222222222

Figure 1: A spread sheet which contains coding of the answers of 4 respondents to the nine-item rating scale survey (n.b. the survey does not contain a Q7 for this data set). Column J of the spread sheet presents the total raw score of each respondent. Column K presents the average response for each respondent.

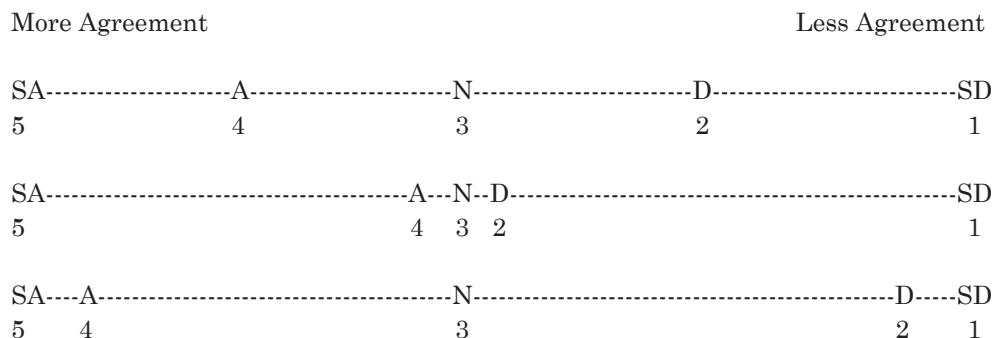


Figure 2: Three different schematics showing the ordinal nature of rating scale data for the same rating scale structure. In the examples there is an ordering of rating scales from Strongly Agree (5) to a Strongly Disagree (1). Of importance is that each rating scale code (e.g. a 3) in this picture shows where a rating scale step might be located, of importance is that one cannot assert or assume the distance between rating scale steps. It is tempting to assume a rating scale functions as shown in the top schematic of figure 2, but this cannot be assumed to be the case. All that is known is the ordering of rating scale steps.

(Strongly Disagree). The bottom line for a researcher is that if one immediately conducts statistical tests with raw data (e.g. the first person in the data set has 29 raw score points, the first person has an average answer of 3.2) one is making a major leap of faith. Unfortunately, statistical tests with such raw data (the 29, the 3.2) can be called into question. This is in part because one does not know the distances between the rating scale categories. Wright and Linacre (1989) provide a more elaborate and detailed discussion of the problematic aspects of evaluating ordinal data as if it were linear data.

Data results must be clearly communicated: A second major problem with commonly used methods of presenting test results (and survey results) is that the results do not often lend them-

selves to communicating the meaning of a respondent's answers to a survey. This means a number can be generated for a respondent (e.g. the 3.2 for the first respondent of figure 1), but one does not have any idea as to what the predicted answer would be for the respondent to each individual survey item. To understand this issue, consider that in this 9-item survey there are some items which are easier to agree to than other survey items. This means the use of a number such as 3.2, to "summarize" a respondent, ignores the fact that not all items are created equal, and as a result this single number (the 3.2) to describe a respondent provides limited context and value to individuals interpreting the survey data.

As readers will note by the conclusion of this introductory article, Rasch techniques help one address the two is-

sues mentioned above, as well as added issues germane to instrument development, use, and communication.

Rasch Steps Which You Can Apply to Your Own Survey Work

Rasch Provides a Framework for Instrument Development

The Rasch model and Rasch theory can be used to provide a strong guiding framework for those developing new instruments (new tests, new surveys) and the framework is just as important for researchers who wish to use existing instruments. The framework is based upon the idea that to “measure” with an instrument (a test of knowledge, an attitude toward something, a reporting about how often one does something) one must measure a single variable, a single trait. This means when one is designing a new instrument, or utilizing an existing instrument, it is critical to understand the importance of only presenting items which help measure a single variable.

There are many instruments in our world that serve as important reminders of the importance of measuring a single variable. Some examples are a voltmeter (to measure voltage), a meter stick (to measure length) and a thermometer (to measure temperature). These devices measure one variable, one trait.

The same must be true of instruments that are used in social science research as well. When a survey is created to measure a trait, for example confidence, it is critical that all the survey items involve the trait of confi-

dence. The survey items measuring confidence should not touch upon other issues. For example, a survey item considering learning might state “I can learn how to add numbers”. That item is very clear, and to the point. The survey item only involves one topic, the topic of confidence in learning. An item that would be problematic would be “I can learn how to write computer code to add numbers”.

Rasch also helps instrument developers (and users) by reminding us that each item of a survey should be viewed as “marking” a part of the trait, the construct. This can be viewed as being similar to the marks of a meter stick which allow one to distinguish between the different lengths of objects. When a survey is created, it is not enough to simply author a set number of survey items. Rather, when respondents have a wide range of attitudes and you simply wish to know where respondents fall on the trait it is important to author survey items which mark different portions of the variable which is to be measured. Too many marks on one part of the trait do not help improve measurement, and too few marks on a part of the trait means one might not be measuring as precisely as one might wish to measure.

For the beginner, it can sometimes be easier to master this idea by first considering an example using test items in which an item can either be correctly answered or not correctly answered. In Table 1 I present a set of 6 math items. Readers will note that there are 5 addition items and 1 division item.

1 + 2 = ?
21 + 3 = ?
31 + 5 = ?
43 + 72 = ?
132 + 434 = ?
9 / 3 = ?

Table 1: A set of 6 potential math items for a test.

From a measurement perspective it is important for the researcher to ask themselves whether it makes sense to include the division item on the test? The bottom line is, what is the variable which the researcher wishes to measure? If the researcher would like to measure a wide range of mathematics skills it might indeed be appropriate to have both addition items and division items on the test. However, an alternative perspective is that if one wishes to measure a respondent's ability with regard to addition, then it makes no sense to include a division test item. Ultimately it is up to the researcher to define what they wish to measure along a single variable, and then to author items (be it for a test or a survey) which involve the single variable. When a researcher does define the variable they wish to measure they should be able to author different items that mark different parts of a trait. The researcher should also be able to make a prediction of which items mark which parts of a trait.

In my experience with scale development, researchers should seek out models which have been proposed for a trait. For example, in the case of measuring respondents' ability with addition, a 5-level model might have been

proposed by experts in the field. And this model could be used to guide the authoring of items for each of the 5 math levels (see Bond (2015) for added information regarding the importance of using a guiding theory for instrument development).

Rasch Provides Plots to Help Researchers Evaluate How Test Items/Survey Items Mark the Trait

Figure 3 presents a schematic of a famous plot used in Rasch measurement, this plot is named a Wright Map (in honor of Benjamin Wright). This Wright Map is organized from the most easy math item to the most difficult math item (some Wright Maps also have respondents and items). The items are located based upon what is named in Rasch measurement "item measures". These item measures express, in the case of a test such as this multiple-choice math test, how difficult the test items are.

For this Wright Map I have chosen to only include the additions items of Table 1. Readers are now urged to review figure 3 and identify which part of the variable line (the black line that extends from the base of the figure to the top of the figure) are marked by test items, and equally important-identify which portions of the trait are not marked by an item.

Readers will be able to see that two items are very close to one another (and thus marking very similar parts of the trait), and readers will be able to see that there are also some gaps between parts of the trait marked by items. It is unbelievable, but prior to the use of Rasch techniques there

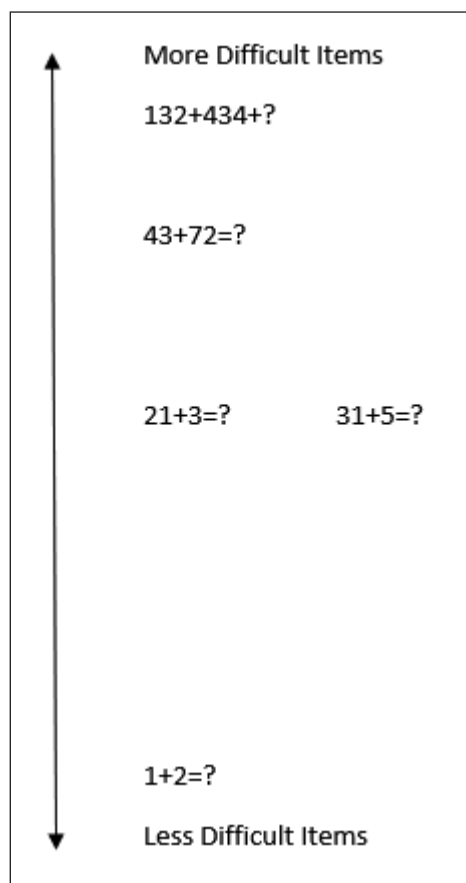


Figure 3: A potential Wright Map following the Rasch analysis of a data set in which item difficulties are computed for each test item. Two items have a similar difficulty.

seems to have been very little attention paid to considering the part of a trait that is marked (and unmarked) by a test item or a survey item.

Why is it important to know how each item of a survey or test mark the variable? As there will always be a limit to the number of items which can be presented to test takers and survey respondents it can be important to not

present items that are redundant (meaning items which mark the same part of the variable). A second issue is that it is important to be aware of where on the meter stick there may be too few (or a total lack of) items marking a trait.

For any test or survey in which a set of items are authored to measure a single trait, it is possible to create such a Wright Map for all items of a survey/test. Then the researcher can identify “gaps” in the items along the trait. To improve the instrument, these gaps should be filled with new items. Doing so can improve the precision of the instrument. Also, a Wright Map can allow one to identify items which are redundant from a measurement perspective (these are the items very near each other in the Wright Map). Generally, an optimal survey or test would be one in which redundant items are removed, and new items authored to fill gaps in the item distribution. The recent text *Rasch Analysis in the Human Sciences* (Boone, Staver & Yale, 2014) provides an entire chapter concerning the topic of Wright Maps. There are of course some added details of Wright Map presentation and interpretation, but these introductory comments should help readers begin to appreciate and use Wright Maps in their own work.

A Sample Rasch Evaluation of Survey Data

For the remainder of this paper I present a basic Rasch evaluation of a survey data set-the data set concerned how often different aspects of language and communication was observed in

preschools. The data was collected from a sample of 300+ respondents. See Rnigsmose & Kragh-Müller (Ring-smose & Kragh-Müller, 2014) for added details regarding the development and use of this instrument. In this brief evaluation a summary of key Rasch observations will be provided as a way of further introducing readers to the application of Rasch measurement. A set of 9 items for a subscale of the survey were:

Q68 There are good opportunities for communication, a good division of adults and children, and a low noise level

Q69 Playing with words is given priority during day to day practice with rhymes and poems

Q70 Singing and song games is part of day to day practice

Q71 Reading practices in which adults and children share their experiences about the book is part of day to day practice

Q72 The children are encouraged to engage in dialogue, the adults encourage the children to engage in dialogue

Q73 There are dialogues and conversations between the children and the adults in which each conversation has “several exchanges” i.e. where several questions are asked and several answers provided by both the adult and the child

Q75 The selection of well-kept games and linguistically stimulation materials is rich and varied

Q76 The furnishings emphasize language stimulation and encouraging conversations thorough meaningful dialogues in play and activities (reading corner)

Q77 The staff initiate activities that motivate the children and encourage communication and contact

The rating scale presented to respondents asked individuals to rate how often a topic (addressed in the items) was observed using a 5-step rating scale. For this article, to simplify plots and tables, recoding was conducted so that all items were expressed with a two-step rating scale in which a “1” indicated a topic took place less often, and a “2” indicated a topic took place more often.

As this is an introductory article, the Rasch evaluation steps to be presented have been selected to be of most use to the beginner who might be interested in utilizing Rasch measurement techniques. For this evaluation the most widely used Rasch software program (Winsteps, Linacre, 2017) was utilized. Table 2 presents part of the Rasch output from the evaluation of the 300+ person data set which resulted from the presentation of the 9-item rating scale survey to respondents.

Spotting Idiosyncratic Items

In the field of Rasch measurement, there are numerous checks that are carried out to investigate whether or

not the items of an instrument, and the rating scale that is being provided to respondents, are functioning as is needed for “measurement” to take place. One of the techniques which is used concerns an investigation of the “fit” of items. This concept can be understood by considering the following example. Imagine that a survey has been developed to measure “self-efficacy” and that item Q2 of the survey is the survey item which is easiest for respondents to “Strongly Agree” with. However, for some reason there are a number of survey respondents who very unexpectedly “Strongly Disagree” with this survey item (these respondents “Strongly Agree” with all other items of the instrument). In this case, it may be that this particular item may have structural issues associated with it (the item may need to be edited in some manner, there may be some unclear wording). Also it could be that the survey developer’s perspective that this item tapped a single variable might be incorrect. It may be that this odd item could be measuring a second (and different variable).

In the table below a key value to investigate is the OUTFIT MNSQ value of each of the instrument items. Generally, one would expect OUTFIT MNSQ values below 1.30 if items are measuring in an expected manner lending themselves to fruitful measurement. In this table it can be quickly seen that item 70 might be an item that is acting in an unusual manner. There are additional steps that are taken to further investigate this item’s functioning, but for the beginner learning about Rasch, the important point is that in Rasch

measurement, one investigates whether or not there is idiosyncratic behavior on the part of each instrument item. In some cases such an investigation helps researchers identify items that might need to be removed from an instrument. Also the identification of such items can help suggest items which might need to be edited, but retained. In fact the same technique and Rasch thinking can be used to investigate the responses of individual survey takers. This means it is also possible to identify respondents who may be answering in a strange manner to the survey items (e.g. perhaps a respondent randomly answering and not reading any of the survey items, perhaps a respondent not understanding the rating scale). If some respondents answer in a very strange manner, one needs to decide what to do with that respondent’s data.

I have found that the value of item 70’s MNSQ Outfit (1.36) probably does not warrant the removal of the item, rather the item should be monitored over time. This means to collect data from a number of samples and evaluate whether or not the item continues to misfit. There are always a limited number of items answered by respondents, and as a result one should be confident that an item is functioning in a manner which does not enhance measurement before removing an item from a survey or test instrument.

Wright and Linacre (1994) provide an overview of this issue. In that work the authors suggest a cutoff of 1.40 Outfit MNSQ for rating scale items. For those interested in further exploring the issue of Fit I suggest Chapter 8

TOTAL SCORE	TOTAL COUNT	MEASURE	S.E.	OUTFIT MNSQ	ITEM
414	304	2.19	.16	1.28	Q68
542	304	-.78	.17	.94	Q69
531	304	-.46	.16	1.36	Q70
465	304	1.05	.15	.96	Q71
543	304	-.81	.17	.85	Q72
525	304	-.31	.16	.87	Q73
496	304	.38	.15	.63	Q75
505	304	.18	.15	.64	Q76
561	304	-1.44	.20	.59	Q77

Table 2: Results of a Rasch evaluation of the survey data using Winsteps (Linacre, 2017). The table provides a number of indices which can be used to evaluate aspects of instrument item functioning. For instance, the value of OUTFIT MNSQ can be used to identify items which might not tap the same trait as the other instrument items. Other headings include Total Score (the total raw score of all respondents to an item), Total Count (the total number of respondents to an item), Measure (the Rasch item measure of an item in units of logits), S. E. (the standard error of an item measure in units of logits), Outfit MNSQ (the outfit of an item) and Item (the survey item).

of Rasch Analysis in the Human Sciences (Boone, Staver & Yale, 2014).

Reliability

Most researchers will be familiar with the term reliability. Very often when a test or survey instrument is created there is the reporting of a “reliability” which ranges from 0 to 1. The most common reliability indices which are provided in articles and talks are an “alpha” or a “KR-20”. In the field of Rasch measurement there are a number of ways in which the reliability of an instrument is investigated. Two ways are through the computation of a Rasch “person reliability” and a Rasch “item reliability”. In this Rasch evaluation the person reliability was .58 and

the item reliability was .97. The important aspect for the Rasch beginner is that with Rasch there are a number of ways in which reliability is evaluated. Two ways in through the computation of a person reliability and an item reliability. Furthermore, when these Rasch reliability calculations are made, the ordinal nature of the rating scale is taken into consideration. Commonly calculations of an alpha or a KR-20 treat ordinal data as if were linear. Linacre (Linacre, 1997) provides some very useful comments regarding this issue. The author of this article believes that Rasch reliabilities can be more informative than traditional reliabilities.

Some Closing Thoughts

It is my hope that the brief Rasch explanations have whet the appetite of readers to consider using Rasch techniques when they wish to develop surveys or tests, and when researchers wish to evaluate the fidelity of instruments (e.g. how well does the instrument work). I will conclude this introduction with a few added comments to help the beginner appreciate why they need to consider the use of Rasch in their work. As I present these examples I wish to stress, that the basics of Rasch are not difficult to learn and apply.

Rasch Provides An Opportunity to Alter Instruments But Still Make Use of Data Collected With A Previous Instrument Form

Often it is the case that when an instrument is developed and utilized for a project- forms of a test or survey are revised (items are edited, added, revised). Rasch techniques allow researchers to remove items from an instrument, add new items to an instrument, all with the goal of improving the measurement functioning of an instrument. However, with Rasch techniques it is possible to express (on the same scale) the performance of respondents regardless of the instrument version which was completed by respondents. This aspect of Rasch is particularly useful in projects where data is collected over time and where there is an interest in improving an instrument to do a better job of measuring. Through the use of Rasch, it is possible to make use of all data collected regardless of instrument version. For in-

stance, one could collect data from respondents with a 10-item survey and also collected data from different respondents with a 15-item survey. And using Rasch techniques it can be possible to express all respondents on the same scale. What is needed is to have some common items in both surveys.

Rasch Provides A Better Way to Communicate Research Findings

Before Rasch, there was really no manner by which a researcher could communicate the meaning of a respondent's performance on a test (which items would one predict a respondent would correctly answer, which items would a respondent be predicted as not being able to correctly answer). Before Rasch that was the same problem for surveys. When a respondent completed a survey, there was no way of communicating what the likely response of a survey taker would be to each item. As survey items differ (in the case of a rating scale with Strongly Agree, Agree, Disagree and Strongly Disagree) in terms of "agreeability", it is critical to be able to express what a respondent would typically have answered for each and every survey item.

Below I provide a Wright Map (Figure 4) created for the 9-item survey which was answered by the 300+ respondents. In this data set a 5-step rating scale was provided to respondents to allow them to indicate how often a surveyed topic (the survey item) took place in a play setting. To simplify the explanation for this introductory article responses were coded as a "1" if something (e.g. singing and song games are part of the day to day prac-

tice) took place less often, and a 2 to indicate something (in this case singing and song games taking place) took place more often.

There are three important aspects of the Wright Map which I would like to call readers attention to: 1) the ordering of items on the right presents a picture of the observed “how often” for each of the survey items. The topic of survey item 77 (Q77 The staff initiate activities that motivate the children and encourage communication and contact) is observed most often, and the topic of survey item 68 (Q68 There are good opportunities for communication, a good division of adults and children, and a low noise level) is observed least often. The ordering of the items from lowest part of the Wright Map to the top of the Wright Map tells a story of what types of play activities are more often observed. Also of interest is that there is a big jump between the location of item 71 (Q71 Reading practices in which adults and children share their experiences about the book is part of day to day practice) and item 68 (There are good opportunities for communication, a good division of adults and children, and a low noise level).

A second aspect of Rasch evaluation is that it is possible to compute the “person measure” for any respondent who completed the survey. The person measures of the sample are marked with the symbols on the left side of the Wright Map.

The third and very important aspect of the Wright Map is that for any respondent (or group of respondents) a

horizontal line can be draw across this Wright Map. On the Wright Map I have marked the location of the average respondent answering the survey (this average person measure is marked with a horizontal line and the letter “M”). Due to the properties of Rasch measurement those item which are plotted below the line are those items for the average respondent where one would predict that the item topic would have taken place at a higher frequency level (a rating of “2” would have been selected). This means that we would predict that typical (average) respondent would have rated items Q75, Q76, Q73, Q70, Q69, Q72, Q77 as occurring frequently (a rating of a “2”). Also those items above the line (Q71, Q68) would be those items for which we would predict the average respondent would report as having the topics taking place less frequently (a “1” from the rating scale). What is amazing for researchers and policy makers is that for any respondent to a survey one can predict how the respondent would have answered each item. Also one can compute mean measures of groups (for example males, females) and plot those mean measures on the Wright Map. And then one can see what it means for females to have an average measure of a specific value, and what it means for males to have an average measure of a specific value. Perhaps most importantly if two lines are drawn, one for each comparison group, the items falling between the lines show which items one group is predicted to be able to answer and not the other group.

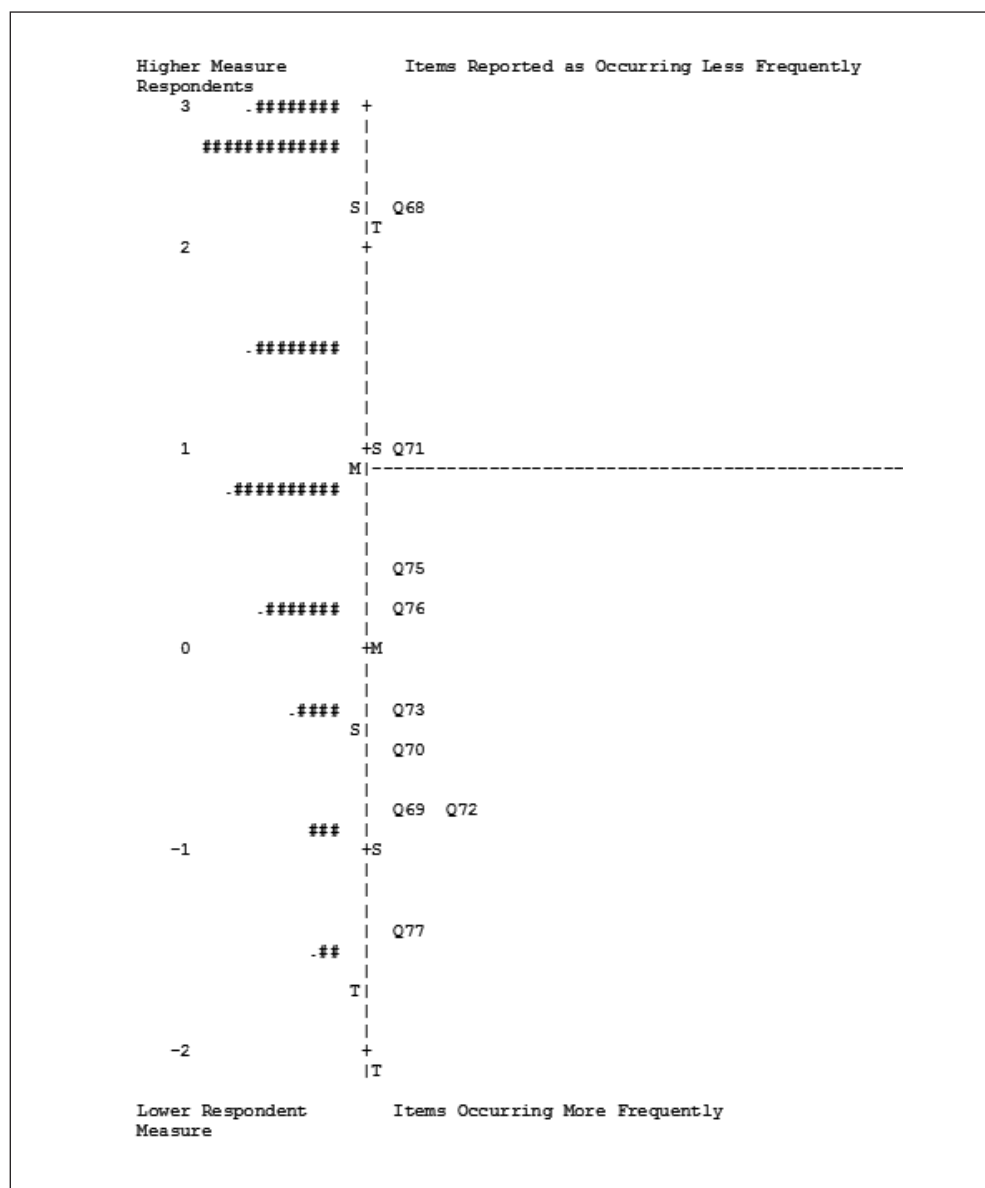


Figure 4: A Wright Map created for the data set. This map has been edited to improve clarity. Respondents are presented on the left side of the Wright Map. Each “#” is 5 respondents, each “.” is 1 to 4 respondents. Respondents with the highest measures are those using the upper end of the rating scale (a “2” for the coding used in this data set), respondents with lower measures are those used fewer ratings of “2” in their answers regarding the frequency with which an item theme is observed. Items are organized from highest measure (item themes observed less often) to items with lower measures (item themes observed more frequently).

Rasch Provides An Opportunity to Link Different Surveys Which Measure the Same Trait

When using Rasch techniques it is possible to link different surveys. Imagine that there were two different self-efficacy instruments (instrument A, instrument B) which were developed by two different researchers. Through a technique which is named “common person linking” it is possible at times to compare a group of respondents who completed instrument A, and a group of respondents who completed instrument B. When the process is completed, it can be possible to express the performance of survey takers on the same scale regardless of which instrument was utilized. The key to linking is having some respondents who complete both forms.

Rasch Provides the Possibility to Evaluate Survey Data With Different Rating Scales

Imagine a confidence survey was answered by respondents. The survey has 10 items which can be answered with Strongly Agree, Agree, Disagree and Strongly Disagree. There are also 4 survey items which can be answered with Yes and No. Also there are 3 survey items which can be answered with Often, Sometimes and Seldom. As long as the 17 survey items all involve the same trait, it is possible to use all survey items to define an overall confidence measure of each respondent.

Rasch Provides an Ability to Evaluate Data Even if Data is Missing

It is not at all uncommon with tests and surveys to have a respondent skip an item, or answer in an unclear manner. Rasch techniques provide the ability to evaluate a data set, even if a respondent has not answered (or unclearly answered) an item. Missing data causes less of a problem in a Rasch evaluation of data than it does with past methods used to evaluate test and survey data.

In Summary

The field of Rasch measurement is one that can be applied in a range of domains in an effort to construct and use measurement instruments such as tests and surveys. Rasch techniques allow instruments of high quality to be constructed, enable person measures to be computed for statistical tests. Tools such as Wright Maps allow research results to be communicated in a useful manner to all who will be interpreting and using the results of a study. I suggest that those wishing to use Rasch to advance their learning by reviewing the introductory text *Rasch Analysis in the Human Sciences* (Boone, Staver & Yale, 2014) and use the widely used Rasch software program Winsteps (Linacre, 2017).

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